ATM Simulator Project Report

Course	Software
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Year	2
Programming Language	Java
Hours	12
Sources	YouTube
Application Type	Console Application

Introduction

The ATM Simulator is a simple banking simulation program developed using Java. The purpose of this project is to demonstrate basic programming concepts such as variables, conditional statements, loops, and user input handling. The system simulates the core operations of an Automated Teller Machine (ATM).

Problem Description

Banks use ATMs to allow customers to perform financial transactions without visiting a branch. This project simulates an ATM that allows the user to check balance, deposit money, and withdraw money while validating the available balance.

System Features

- Check account balance.
- Deposit money into the account.
- Withdraw money with balance validation.
- Simple user menu for interaction.

Code Explanation

The program starts from the main method where a menu is displayed to the user. The user selects an operation, and the program executes the corresponding function. Conditional statements are used to control the flow, while variables store the current balance and transaction amounts.

Screenshots

```
=== ATM MENU ===
1. Check Balance
2. Withdraw
3. Deposit
4. Exit
Enter your choice: 1
Your balance is: 1000.0

=== ATM MENU ===
1. Check Balance
2. Withdraw
3. Deposit
4. Exit
Enter your choice:
```

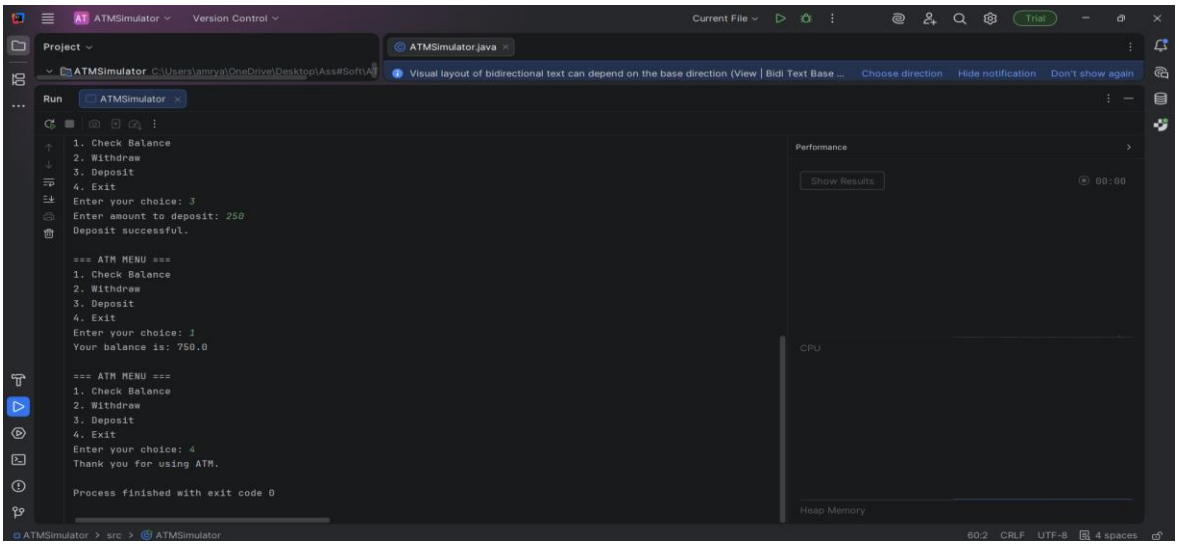
ATMSimulator.java

```java
import java.util.Scanner;

public class ATMSimulator {

    public static void main(String[] args) {

        Scanner input = new Scanner(System.in);

        double balance = 1000; // الرصيد الابتدائي
        int choice;

        do {
            System.out.println("\n=== ATM MENU ===");
            System.out.println("1. Check Balance");
            System.out.println("2. Withdraw");
            System.out.println("3. Deposit");
            System.out.println("4. Exit");
            System.out.print("Enter your choice: ");

            choice = input.nextInt();

            switch (choice) {

                case 1:
                    System.out.println("Your balance is: " + balance);
                    break;

                case 2:
                    System.out.print("Enter amount to withdraw: ");
                    double withdraw = input.nextDouble();
```

```
=== ATM MENU ===
1. Check Balance
2. Withdraw
3. Deposit
4. Exit
Enter your choice: 2
Invalid choice!

=== ATM MENU ===
1. Check Balance
2. Withdraw
3. Deposit
4. Exit
Enter your choice: 9
Invalid choice!

=== ATM MENU ===
1. Check Balance
2. Withdraw
3. Deposit
4. Exit
Enter your choice: 1
Your balance is: 1000.0
```

```
Your balance is: 1000.0

=== ATM MENU ===
1. Check Balance
2. Withdraw
3. Deposit
4. Exit
Enter your choice: 2
Enter amount to withdraw: 500
Withdrawal successful.

=== ATM MENU ===
1. Check Balance
2. Withdraw
3. Deposit
4. Exit
Enter your choice: 1
Your balance is: 500.0

=== ATM MENU ===
1. Check Balance
2. Withdraw
3. Deposit
4. Exit
Enter your choice:
```

```
Your balance is: 500.0

=== ATM MENU ===
1. Check Balance
2. Withdraw
3. Deposit
4. Exit
Enter your choice: 3
Enter amount to deposit: 250
Deposit successful.

=== ATM MENU ===
1. Check Balance
2. Withdraw
3. Deposit
4. Exit
Enter your choice: 1
Your balance is: 750.0

=== ATM MENU ===
1. Check Balance
2. Withdraw
3. Deposit
4. Exit
Enter your choice:
```



Main Function

Component	Type / Name	Description
Input Scanner	Scanner input	Creates an object to read user input from the console.
Initial Balance	double balance	Sets the starting funds (defaulted to 1000).
User Choice	int choice	Stores the menu option selected by the user ($1-4$).
Loop Type	do-while	Ensures the menu displays at least once and continues until the user chooses "Exit" (4).
Decision Logic	switch (choice)	Directs the program to specific code blocks based on the user's input.
Validation	if (withdraw <= balance)	Checks if the user has enough money before allowing a withdrawal.
Cleanup	input.close()	Properly closes the scanner to prevent memory leaks.

Link of Video → https://drive.google.com/file/d/1eS0M8JXLTs6ysetBm6V84VCBZ0yad6NP/view?usp=drive_link